

Introduction to PBGoF

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Installation

Install PBGoF from GitHub with devtools:

```
if (!"devtools" %in% rownames(installed.packages())) {  
  install.packages("devtools")  
}  
devtools::install_github("Divo-Lee/PBGoF")
```

Alternatively, install it with pak:

```
if (!"pak" %in% rownames(installed.packages())) {  
  install.packages("pak")  
}  
pak::pkg_install("Divo-Lee/PBGoF")
```

After installation, load the package with:

```
library(PBGoF)
```

Overview

PBGoF provides goodness-of-fit tests for assessing whether a numeric sample is compatible with a univariate skew-normal (SN) distribution when its parameters are unknown and estimated from the same data. This is a composite goodness-of-fit problem: estimating the location, scale, and shape parameters changes the null distribution of standard empirical distribution function (EDF) statistics. Consequently, ordinary Kolmogorov–Smirnov p-values for a completely specified distribution are not appropriate.

The package offers two complementary approaches:

1. **Parametric bootstrap tests** simulate a reference distribution for the observed sample and re-estimate all parameters in every replicate.
2. **Precomputed-quantile tests** use tables obtained from 100,000 Monte Carlo replicates for each combination of sample size and centered skewness represented in the bundled tables.

Both approaches are available for the Kolmogorov–Smirnov (KS) and Cramér–von Mises (CvM) statistics. Parameter estimation is performed by `sn.fit.robust()`, which uses a sequence of increasingly stabilized fitting methods.

The skew-normal model

Under the direct parameterization (DP), the skew-normal model has location ξ , positive scale ω , and shape α . Setting α to zero gives a normal distribution; positive and negative values produce right- and left-skewed densities, respectively.

The centered parameterization (CP) expresses the same model through:

- mean: the population mean;
- sd: the population standard deviation;
- gamma1: the population coefficient of skewness.

DP is convenient for evaluating the fitted SN distribution function. CP is convenient for matching fitted skewness to the precomputed simulation tables. The signs of alpha and gamma1 agree: positive alpha corresponds to positive gamma1, and negative alpha corresponds to negative gamma1.

Why lookup uses the absolute skewness

Suppose X follows an SN distribution with DP parameters (ξ, ω, α) . The reflected variable $-X$ is also skew-normal, with shape $-\alpha$. Reflection reverses the sign of the CP skewness gamma1 but preserves the magnitude and does not change the sampling distribution of a reflection-invariant EDF goodness-of-fit statistic.

Mateu-Figueras, Puig, and Pewsey (2007) explicitly report that the distributions of the five EDF statistics they study are invariant to changes in the sign of the skew-normal shape parameter. Their test instructions state that when the fitted shape parameter is negative, the table corresponding to its positive counterpart should be used.

PBGoF indexes its simulation tables by the CP coefficient gamma1 rather than the DP shape alpha. Because gamma1 changes sign together with alpha, PBGoF implements the same symmetry through

```
gamma1_used <- round(abs(gamma1_hat), 2)
```

followed by restriction to the available table range 0.01–0.99. Thus, for example, fitted values gamma1_hat = -0.63 and gamma1_hat = 0.63 both use the gamma1 = 0.63 row. The original sign is retained in gamma1_hat in the returned result, while gamma1_used records the non-negative lookup value.

Robust parameter estimation

```
library(PBGoF)
```

```
set.seed(2026)
```

```
x <- sn::rsn(100, xi = 0, omega = 1, alpha = 4)
```

```
sn.fit.robust(x, para_form = "DP")
```

```
sn.fit.robust(x, para_form = "CP")
```

sn.fit.robust() first attempts ordinary maximum likelihood estimation (MLE). If the fit does not yield finite estimates and standard errors, it tries maximum penalized likelihood estimation (MPLE) with the default penalty and then MPLE with the matching-prior penalty. If every attempt fails, the function returns a named vector of missing values and issues a warning.

Here, *robust* refers to protection against numerical fitting failures. It does not mean that the estimator is resistant to outliers or contamination. Users should still inspect their data and assess whether the skew-normal family is a scientifically reasonable model.

The input must be a numeric vector containing at least 10 finite observations and at least two distinct values. Matrices, factors, missing values, and infinite values are rejected rather than silently converted.

Graphical assessment of the fitted model

sn.plot.check() provides a quick visual comparison between the observed sample and its fitted skew-normal distribution. It draws a density-scale histogram, adds the fitted SN density, and optionally marks individual observations with a rug plot.

The following example generates a sample directly with the `sn` package and then checks the fitted model:

```
set.seed(123)
x_plot <- sn::rsn(
  n = 200,
  xi = 1,
  omega = 2,
  alpha = 5
)
```

```
plot_result <- sn.plot.check(x_plot)
plot_result$parameters
```

The function fits the model through `sn.fit.robust()`, so its parameter estimates use the same numerical fallback strategy as the formal PGoF tests. It returns the fitted DP parameters and plotted coordinates invisibly, allowing the underlying values to be inspected or tested programmatically.

Visual inspection can reveal features that a single p-value does not describe, including isolated outliers, multimodality, tail discrepancies, and systematic differences between the histogram and fitted density. The appearance of a histogram depends on its breaks, so it is often useful to compare several choices:

```
sn.plot.check(x_plot, breaks = "FD")
sn.plot.check(x_plot, breaks = "Scott")
sn.plot.check(x_plot, breaks = 20)
```

The plot is a diagnostic aid rather than a formal decision rule. It should be used alongside `PGoF_ks_test()`, `PGoF_cvm_test()`, or their parametric bootstrap counterparts.

Parametric bootstrap tests

The bootstrap functions are:

```
sn.para.bootstrap.ks.test(x, B = 1000, seed = 103)
sn.para.bootstrap.cvm.test(x, B = 1000, seed = 103)
```

For each test, PGoF performs these steps:

1. Fit an SN distribution to the observed data.
2. Calculate the observed EDF statistic using the fitted DP parameters.
3. Generate B samples of the same size from the fitted SN distribution.
4. Refit the SN distribution separately in every bootstrap sample.
5. Recalculate the statistic using each bootstrap sample's fitted parameters.
6. Estimate the p-value from the proportion of valid bootstrap statistics at least as large as the observed statistic.

The finite-simulation correction adds one to both the exceedance count and the number of valid replicates. It prevents a p-value of exactly zero. Failed fits are excluded. The returned numeric p-value has attributes recording the observed statistic, requested number of replicates, and numbers of valid and failed replicates:

```
p <- sn.para.bootstrap.ks.test(x, B = 999, seed = 103)
p
attributes(p)
```

The seed argument makes the bootstrap reproducible. PGoF restores the caller's previous random-number state when the test finishes. Set `seed = NULL` to continue from the current random-number stream.

Larger values of B give finer and more stable p-values but require more computation. Values such as 999 or 1999 are useful during analysis; substantially larger values may be preferable for final inference near a decision threshold.

Tests based on precomputed quantiles

The fast lookup functions are:

```
PBGoF_ks_test(x)
PBGoF_cvm_test(x)
```

They avoid running a new bootstrap for each dataset. For observed data, the functions:

1. estimate DP parameters for evaluating the fitted distribution;
2. estimate CP parameters and calculate $\text{abs}(\text{gamma1})$, so negative and positive fitted skewness of the same magnitude use the same reference distribution;
3. round the absolute skewness to two decimal places and restrict it to the table range 0.01–0.99;
4. select the table row for the sample size and matched skewness;
5. compare the observed statistic with the stored 0.01–0.99 quantiles.

The bundled tables cover sample sizes 30 through 500. For a sample larger than 500, all observations are retained when fitting the model and constructing the EDF, but the lookup sample size and external statistic scaling are set to 500. The result reports both values:

```
set.seed(1)
x_large <- sn::rsn(750, alpha = 3)
result <- PBGoF_ks_test(x_large)

result$n          # 750: actual number of observations
result$n_used     # 500: scaling and lookup value
```

Mateu-Figueras, Puig, and Pewsey (2007) reported that the quantiles of the EDF statistics for sample sizes above 500 were almost identical to those for $n = 500$, and recommended using the $n = 500$ critical values. Replacing the external scaling value by 500 is the PBGoF convention used to remain consistent with the construction of its bundled tables; the full empirical distribution always uses all observations.

The lookup result contains:

- statistic: observed scaled test statistic;
- n: actual sample size;
- n_used: sample size used for scaling and table lookup;
- gamma1_hat: fitted signed CP skewness;
- gamma1_used: rounded absolute skewness used for lookup;
- p.value: approximate upper-tail p-value.

Because the stored probability grid advances in increments of 0.01, lookup p-values are conservative step-function approximations restricted to 0.01–0.99. They should not be interpreted as having greater precision than the table permits.

Advanced users can supply a compatible custom table:

```
PBGoF_ks_test(x, ks_table = my_ks_table)
PBGoF_cvm_test(x, cvm_table = my_cvm_table)
```

A custom table must be a data frame containing `n`, `gamma1`, and probability columns named `q_0.01` through `q_0.99`. Quantiles within a selected row must be finite and nondecreasing.

Choosing between the approaches

Use the **parametric bootstrap** when maximum fidelity to the observed sample size is important, when the sample size is below the table range, or when a more finely resolved p-value is needed. Its main cost is computation because every bootstrap sample must be refitted.

Use a **precomputed-quantile test** for rapid repeated screening when the table’s sample-size and skewness grid is appropriate. It is especially useful when many datasets must be tested, but its p-values are discretized and depend on the simulation design used to create the tables.

For important conclusions, it can be useful to run the lookup test first and confirm borderline results with a sufficiently large parametric bootstrap.

Interpretation and limitations

A small p-value indicates that the observed EDF discrepancy is unusually large under the fitted skew-normal model. It is evidence against that model, not a measure of the practical importance of the discrepancy. Conversely, a large p-value does not prove that the data are skew-normal; it means that the test did not detect a departure at the available sample size and resolution.

The tests assume independent observations from a continuous distribution. Serial dependence, clustering, censoring, rounding, or many ties can alter the null distribution. The procedures also inherit limitations of skew-normal parameter estimation, especially near symmetry or extreme skewness. Graphical checks such as histograms, fitted densities, and probability plots should be used alongside the formal tests.

References

- Azzalini, A. (1985). A class of distributions which includes the normal ones. *Scandinavian Journal of Statistics*, **12**(2), 171–178.
- Azzalini, A., and Capitanio, A. (2014). *The Skew-Normal and Related Families*. Cambridge University Press.
- Babu, G. J., and Rao, C. R. (2004). Goodness-of-fit tests when parameters are estimated. *Sankhya*, **66**, 63–74.
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